PREFACE

In the realm of artificial intelligence, the fusion of visual perception and natural language understanding heralds a new era of interaction between humans and machines. This report embarks on an exploration of Visual Question Answering (VQA), a captivating domain where algorithms are tasked with deciphering and responding to inquiries about visual content.

Drawing upon the latest advancements in deep learning and natural language processing, our VQA system embodies a synthesis of cutting-edge techniques. Leveraging convolutional neural networks (CNNs) for image analysis, transformer-based models like BERT for textual comprehension, and attention mechanisms for context-aware reasoning, our endeavour embodies the convergence of vision and language modalities.

Beyond mere technical exposition, this report delves into the theoretical underpinnings that drive the efficacy of our VQA system. From the elucidation of attention mechanisms to the nuances of transfer learning and image region detection, each concept is dissected to provide a deeper understanding of our methodology.